2. Online Hotel Room Booking Website

Aim

To design and develop a responsive online hotel room booking website that allows users to register, search for available rooms, book rooms, manage bookings, and view booking history. It also allows administrators to manage room types, prices, availability, bookings, and website information.

CODE

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Hotel Room Booking System</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      background: #f4f7fa;
      color: #333;
    }

    header, footer {
      background: #17365d;
      color: white;
      text-align: center;
      padding: 20px;
    }

    nav {
      background: #24527a;
      padding: 12px;
      text-align: center;
    }

    nav a {
      color: white;
```

```
    margin: 10px;
    text-decoration: none;
}

section {
    padding: 25px;
}

h2 {
    color: #17365d;
}

.container {
    max-width: 850px;
    margin: auto;
    background: white;
    padding: 20px;
    border-radius: 8px;
}

input, select, textarea, button {
    width: 100%;
    padding: 10px;
    margin: 8px 0;
    box-sizing: border-box;
}

button {
    background: #17365d;
    color: white;
    border: none;
    cursor: pointer;
}
```

```
    button:hover {
      background: #2874a6;
    }

    .rooms {
      display: grid;
      grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
      gap: 15px;
    }

    .room {
      background: white;
      padding: 18px;
      border-radius: 8px;
      box-shadow: 0 2px 8px #ddd;
    }

    .room h3 {
      color: #2874a6;
    }

    .result {
      margin-top: 15px;
      font-weight: bold;
    }
  </style>
</head>
<body>

<header>
  <h1>Welcome to Grand Stay Hotel</h1>
  <p>Comfortable Rooms | Easy Booking | Best Service</p>
</header>
```

```
<nav>
  <a href="#home">Home</a>
  <a href="#rooms">Rooms</a>
  <a href="#register">Register</a>
  <a href="#booking">Book Room</a>
  <a href="#history">Booking History</a>
  <a href="#admin">Admin</a>
  <a href="#contact">Contact</a>
</nav>

<section id="home">
  <div class="container">
    <h2>About Us</h2>
    <p>Grand Stay Hotel provides comfortable accommodation
      for families, tourists and business travellers.</p>

    <h3>Room Types</h3>
    <p>Standard Room, Deluxe Room, Suite Room and Family Room.</p>
  </div>
</section>

<section id="rooms">
  <h2 style="text-align:center;">Available Room Types</h2>
  <div class="rooms" id="roomList"></div>
</section>

<section id="register">
  <div class="container">
    <h2>User Registration / Login</h2>

    <form id="registerForm">
      <input type="text" id="userName"
          placeholder="Full Name" required>
      <input type="email" id="userEmail"
```

```
            placeholder="Email Address" required>
        <input type="password" id="userPassword"
            placeholder="Password" required>
        <button type="submit">Register</button>
      </form>

      <form id="loginForm">
        <h3>User Login</h3>
        <input type="email" id="loginEmail"
            placeholder="Email Address" required>
        <input type="password" id="loginPassword"
            placeholder="Password" required>
        <button type="submit">Login</button>
      </form>

      <div id="profile"></div>
    </div>
  </section>

  <section id="booking">
    <div class="container">
      <h2>Book a Room</h2>

      <form id="bookingForm">
        <input type="text" id="guestName"
            placeholder="Guest Name" required>

        <select id="roomType" required>
          <option value="">Select Room Type</option>
          <option>Standard Room</option>
          <option>Deluxe Room</option>
          <option>Suite Room</option>
          <option>Family Room</option>
        </select>
```

```
        <label>Check-in Date</label>
        <input type="date" id="checkIn" required>

        <label>Check-out Date</label>
        <input type="date" id="checkOut" required>

        <input type="number" id="guests"
            min="1" max="10" placeholder="Number of Guests"
            required>

        <button type="button" onclick="checkAvailability()">
          Check Availability
        </button>

        <div id="availability" class="result"></div>

        <button type="submit">Confirm Booking</button>
      </form>
    </div>
</section>

<section id="history">
    <div class="container">
      <h2>Booking History</h2>
      <div id="bookingHistory">No bookings available.</div>
    </div>
</section>

<section id="admin">
    <div class="container">
      <h2>Administrator Panel</h2>

      <form id="adminLogin">
```

```
        <input type="password" id="adminPassword"
            placeholder="Demo Admin Password" required>
        <button type="submit">Admin Login</button>
      </form>

      <div id="adminPanel" style="display:none;">
        <h3>Manage Room Quantity and Price</h3>

        <select id="adminRoom">
          <option>Standard Room</option>
          <option>Deluxe Room</option>
          <option>Suite Room</option>
          <option>Family Room</option>
        </select>

        <input type="number" id="roomPrice"
            placeholder="Price per Night" min="1">
        <input type="number" id="roomQuantity"
            placeholder="Available Rooms" min="0">

        <button onclick="updateRoom()">Update Room</button>

        <h3>Manage Bookings</h3>
        <div id="adminBookings"></div>

        <h3>Update Website Information</h3>
        <textarea id="pageInfo" rows="4"
            placeholder="Enter hotel information"></textarea>
        <button onclick="updateInformation()">Update Information</button>

        <div id="adminMessage"></div>
      </div>
    </div>
  </section>
```

```
<section id="contact">
  <div class="container">
    <h2>Contact Us</h2>
    <p>Email: info@grandstay.example</p>
    <p>Phone: +91 98765 43210</p>
    <p>Address: Main Road, City Centre</p>

    <h3>Customer Service Q&A</h3>
    <p><b>Q: Can I cancel a booking?</b></p>
    <p>A: Demo bookings can be cancelled at least three days before check-in.</p>

    <p><b>Q: Can I change my booking?</b></p>
    <p>A: Demo bookings can be modified at least three days before check-in.</p>

    <h3>Local Travel Guide</h3>
    <p>Ask the hotel reception for nearby attractions, transport,
      restaurants and local travel information.</p>

    <h3>Privacy Policy</h3>
    <p>Customer information should be collected and handled securely.
      This demonstration does not provide secure server-side storage.</p>
  </div>
</section>

<footer>
  <p>&copy; 2026 Grand Stay Hotel. All Rights Reserved.</p>
</footer>

<script>
  let rooms = [
    {name: "Standard Room", price: 1500, quantity: 10},
    {name: "Deluxe Room", price: 2500, quantity: 8},
    {name: "Suite Room", price: 4000, quantity: 5},
```

```
  {name: "Family Room", price: 3500, quantity: 6}
];

let users = [];
let bookings = [];
let currentUser = null;
let nextBookingId = 1;

function renderRooms() {
  document.getElementById("roomList").innerHTML = rooms.map(room => `
    <div class="room">
      <h3>${room.name}</h3>
      <p>Price per night: ₹${room.price}</p>
      <p>Rooms available: ${room.quantity}</p>
      <p>Free Wi-Fi, clean rooms and room service.</p>
    </div>
  `).join("");
}

document.getElementById("registerForm").addEventListener("submit", function(e) {
  e.preventDefault();

  const name = document.getElementById("userName").value.trim();
  const email = document.getElementById("userEmail").value.trim();
  const password = document.getElementById("userPassword").value;

  if (users.some(user => user.email === email)) {
    alert("This email is already registered.");
    return;
  }

  users.push({name, email, password});
  alert("Registration successful. Please log in.");
  this.reset();
```

```
});

document.getElementById("loginForm").addEventListener("submit", function(e) {
  e.preventDefault();

  const email = document.getElementById("loginEmail").value.trim();
  const password = document.getElementById("loginPassword").value;

  const user = users.find(u => u.email === email && u.password === password);

  if (!user) {
    alert("Invalid login details. Register first.");
    return;
  }

  currentUser = user;
  document.getElementById("profile").textContent =
    "Welcome, " + user.name + "!";
  document.getElementById("guestName").value = user.name;
  showHistory();
});

function checkAvailability() {
  const roomName = document.getElementById("roomType").value;
  const start = document.getElementById("checkIn").value;
  const end = document.getElementById("checkOut").value;

  const result = document.getElementById("availability");

  if (!roomName || !start || !end) {
    result.textContent = "Select a room and both dates.";
    return;
  }
```

```
  if (end <= start) {
    result.textContent = "Check-out must be after check-in.";
    return;
  }

  const room = rooms.find(r => r.name === roomName);
  const overlapping = bookings.filter(b =>
    b.roomName === roomName &&
    b.status === "Confirmed" &&
    start < b.checkOut &&
    end > b.checkIn
  ).length;

  const available = room.quantity - overlapping;

  result.textContent = available > 0
    ? available + " room(s) available for your selected dates."
    : "No rooms available for those dates.";
}

document.getElementById("bookingForm").addEventListener("submit", function(e) {
  e.preventDefault();

  if (!currentUser) {
    alert("Please register and log in before booking.");
    return;
  }

  const roomName = document.getElementById("roomType").value;
  const checkIn = document.getElementById("checkIn").value;
  const checkOut = document.getElementById("checkOut").value;
  const guests = Number(document.getElementById("guests").value);
  const room = rooms.find(r => r.name === roomName);
```

```
if (!room || !checkIn || !checkOut || checkOut <= checkIn) {
  alert("Please enter valid booking details.");
  return;
}

const overlapping = bookings.filter(b =>
  b.roomName === roomName &&
  b.status === "Confirmed" &&
  checkIn < b.checkOut &&
  checkOut > b.checkIn
).length;

if (overlapping >= room.quantity) {
  alert("Sorry, no rooms are available for those dates.");
  return;
}

const nights = Math.round(
  (new Date(checkOut) - new Date(checkIn)) / 86400000
);

const booking = {
  id: nextBookingId++,
  email: currentUser.email,
  guestName: document.getElementById("guestName").value,
  roomName,
  checkIn,
  checkOut,
  guests,
  total: nights * room.price,
  status: "Confirmed"
};

bookings.push(booking);
```

```
  alert("Demo booking confirmed! Total: ₹" + booking.total);
  showHistory();
  showAdminBookings();
  this.reset();
});

function showHistory() {
  const history = document.getElementById("bookingHistory");

  if (!currentUser) {
    history.textContent = "Please log in to view your booking history.";
    return;
  }

  const mine = bookings.filter(b => b.email === currentUser.email);

  history.innerHTML = mine.length ? mine.map(b => `
    <div class="room">
      <p>Booking ID: ${b.id}</p>
      <p>Room: ${b.roomName}</p>
      <p>Dates: ${b.checkIn} to ${b.checkOut}</p>
      <p>Total: ₹${b.total}</p>
      <p>Status: ${b.status}</p>
      ${b.status === "Confirmed" ? `
        <button onclick="cancelBooking(${b.id})">Cancel Booking</button>
        <button onclick="modifyBooking(${b.id})">Modify Dates</button>
      ` : ""}
    </div>
  `).join("") : "No bookings available.";
}

function cancelBooking(id) {
  const booking = bookings.find(b => b.id === id);
```

```
  if (!booking || booking.email !== currentUser.email) return;

  const days = (new Date(booking.checkIn) - new Date()) / 86400000;

  if (days < 3) {
    alert("Bookings can only be cancelled at least 3 days before arrival.");
    return;
  }

  booking.status = "Cancelled";
  showHistory();
  showAdminBookings();
}

function modifyBooking(id) {
  const booking = bookings.find(b => b.id === id);

  if (!booking || booking.email !== currentUser.email) return;

  const days = (new Date(booking.checkIn) - new Date()) / 86400000;

  if (days < 3) {
    alert("Bookings can only be modified at least 3 days before arrival.");
    return;
  }

  const newCheckIn = prompt("Enter new check-in date (YYYY-MM-DD):", booking.checkIn);
  const newCheckOut = prompt("Enter new check-out date (YYYY-MM-DD):", booking.checkOut);

  if (!newCheckIn || !newCheckOut || newCheckOut <= newCheckIn) {
    alert("Invalid dates.");
    return;
  }
```

```
    booking.checkIn = newCheckIn;
    booking.checkOut = newCheckOut;

    const room = rooms.find(r => r.name === booking.roomName);
    const nights = (new Date(newCheckOut) - new Date(newCheckIn)) / 86400000;
    booking.total = nights * room.price;

    showHistory();
    showAdminBookings();
}

document.getElementById("adminLogin").addEventListener("submit", function(e) {
    e.preventDefault();

    // Demonstration only. Real admin authentication needs a backend.
    if (document.getElementById("adminPassword").value === "admin123") {
        document.getElementById("adminPanel").style.display = "block";
        showAdminBookings();
        alert("Demo admin login successful.");
    } else {
        alert("Incorrect demo admin password.");
    }
});

function updateRoom() {
    const name = document.getElementById("adminRoom").value;
    const price = Number(document.getElementById("roomPrice").value);
    const quantity = Number(document.getElementById("roomQuantity").value);

    if (price <= 0 || quantity < 0 ||
        document.getElementById("roomPrice").value === "" ||
        document.getElementById("roomQuantity").value === "") {
        alert("Enter a valid price and room quantity.");
```

```
    return;
  }

  const room = rooms.find(r => r.name === name);
  room.price = price;
  room.quantity = quantity;

  renderRooms();
  showAdminBookings();

  document.getElementById("adminMessage").textContent =
    "Room details updated in this demo.";
}

function showAdminBookings() {
  const area = document.getElementById("adminBookings");

  if (!bookings.length) {
    area.textContent = "No bookings yet.";
    return;
  }

  area.innerHTML = bookings.map(b => `
    <div class="room">
      <p>Booking ID: ${b.id}</p>
      <p>Customer: ${b.guestName} (${b.email})</p>
      <p>Room: ${b.roomName}</p>
      <p>Dates: ${b.checkIn} to ${b.checkOut}</p>
      <p>Status: ${b.status}</p>
      <button onclick="adminCancel(${b.id})">Cancel Booking</button>
      <button onclick="adminModify(${b.id})">Change Dates</button>
    </div>
  `).join("");
}
```

```
function adminCancel(id) {
  const booking = bookings.find(b => b.id === id);
  if (!booking) return;

  booking.status = "Cancelled";
  showAdminBookings();
  showHistory();
}

function adminModify(id) {
  const booking = bookings.find(b => b.id === id);
  if (!booking) return;

  const checkIn = prompt("New check-in date:", booking.checkIn);
  const checkOut = prompt("New check-out date:", booking.checkOut);

  if (!checkIn || !checkOut || checkOut <= checkIn) {
    alert("Invalid dates.");
    return;
  }

  booking.checkIn = checkIn;
  booking.checkOut = checkOut;

  const room = rooms.find(r => r.name === booking.roomName);
  booking.total =
    ((new Date(checkOut) - new Date(checkIn)) / 86400000) * room.price;

  showAdminBookings();
  showHistory();
}

function updateInformation() {
```

```
    const text = document.getElementById("pageInfo").value.trim();

    if (!text) {
      alert("Enter the information you want to update.");
      return;
    }

    alert("Information entered. Saving it permanently requires a backend.");
  }

  renderRooms();
</script>

</body>
</html>
```

